Please write clearly in block capitals. Centre number | Candidate number

Surname

Forename(s)

A-level

ENVIRONMENTAL SCIENCE

Paper 2 (Generated practice paper - Set 24)

Assessed: The living environment, Biological resources, Sustainability, Research methods

Time allowed: 3 hours

Materials

For this paper you may use:
- a calculator.

Instructions

- Use black ink or black ball-point pen. Pencil should only be used for drawing.
- Fill in the boxes at the top of this page.
- Answer **all** questions 1 to 10 and **one** essay from question 11.
- You must answer the questions in the spaces provided. Do not write outside the box around each page or on blank pages.
- Do all rough work in this book. Cross through any work you do not want to be marked.

Information

- The marks for questions are shown in brackets.
- The maximum mark for this paper is 120.
- All questions should be answered in continuous prose.
- You will be assessed on your ability to:
  - use good English
  - organise information clearly
  - use specialist vocabulary where appropriate.

For Examiner's Use

Question	Mark
1	
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10	
11	
TOTAL	

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Answer all questions in the spaces provided.

0 1 The Rio Bravo Conservation and Management Area in Belize exploits mahogany under a sustainable management plan while conserving a rainforest of over 200 tree, 400 bird and 70 mammal species.

0 1 . 1 Describe **five** features of the timber harvesting at Rio Bravo that make it sustainable.

[5 marks]

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0 1 . 2 Explain why involving local communities is essential to the success of the reserve.

[3 marks]

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0 1 . 3 Explain why selective logging is better for wildlife than clear-felling, and give **one** disadvantage.

[2 marks]

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0 2 . 1 Explain how overfishing of the giant triton mollusc damaged coral reefs in Australia.

[2 marks]

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0 2 . 2 Describe **four** management practices used in marine protected areas to protect coral reefs.

[4 marks]

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0 2 . 3 Explain why deep-water coral reefs recover much more slowly from trawling damage than tropical reefs.

[2 marks]

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0 2 . 4 Give **two** threats to deep-water reefs and **one** conservation measure.

[2 marks]

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0 3 . 1 Explain how each of the following changes in abiotic factors caused by human activities can make a habitat unsuitable for species living there.

[6 marks]

Water availability

Dissolved oxygen

pH

0 3 . 2 Explain how increased water turbidity caused by ploughing, mining or dredging harms aquatic organisms.

[2 marks]

0 3 . 3 Explain why species that were already rare are most likely to be lost when abiotic conditions change.

[2 marks]

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0 4 . 1 Explain why electronic meters must be calibrated before use and describe how a pH meter is calibrated.

[3 marks]

0 4 . 2 Describe how wind velocity measurements with an anemometer should be standardised.

[3 marks]

0 4 . 3 Describe how a whirling hygrometer is used to measure humidity.

[2 marks]

0 4 . 4 Describe **two** methods of measuring nitrate concentration in water.

[2 marks]

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0 5 . 1 Which of the following would be most suitable for monitoring the movements of a whale over several months? Tick one box.

[1 mark]

A Radio collar ☐

B Acoustic / satellite tag ☐

C Pitfall trap ☐

D Geolocator tag recovered by recapture ☐

0 5 . 2 Explain why post-release monitoring is important in a captive breeding and release programme and describe **two** suitable methods.

[4 marks]

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0 6 . 1 Explain how the development of biogeochemical cycles and of transpiration by land plants helped to maintain stable conditions for life.

[3 marks]

0 6 . 2 Describe **three** limitations of early methods of studying past conditions on Earth.

[3 marks]

1

2

3

0 6 . 3 Describe how each of the following modern techniques provides information about past conditions.

[4 marks]

Isotope analysis of ice cores

Gas analysis of ice cores

Radiocarbon dating

Magnesium : calcium ratio in calcite

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0 7 . 1 Which of the following is a feature of natural systems? Tick **one** box.

[1 mark]

A Reactions that require high temperatures ☐

B Reliance on non-renewable energy ☐

C Waste products that are the raw materials for other processes ☐

D Dispersal of elements so they cannot be re-used ☐

0 7 . 2 Explain why nitrogen fixation by bacteria is more sustainable than the Haber process.

[2 marks]

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0 7 . 3 Explain why incineration of wastes is less sustainable than natural decomposition.

[2 marks]

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0 8 . 1 Explain what a carbon footprint should include when calculated for a car.

[3 marks]

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0 8 . 2 Describe what the WWF Living Planet Index measures and how it is used.

[2 marks]

0 8 . 3 Describe how the Ol Pejeta reserve in Kenya combines commercial cattle ranching with wildlife conservation.

[5 marks]

10

0 9 . 1 Explain why food surpluses produced in MEDCs in the 1970s could not easily be sold.

[3 marks]

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0 9 . 2 Describe **four** methods used to reduce over-production while maintaining farm incomes.

[4 marks]

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0 9 . 3 Describe **three** features that farmers may be paid for under an agri-environmental scheme and the conservation purpose of each.

[3 marks]

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1 0 Table 1 shows the catch made by a shrimp trawler during one fishing trip.

Table 1

Component of catch	Mass / tonnes
Shrimps (target species)	0.6
Crabs, sea urchins, molluscs and starfish	5.9
Immature fish of commercial species	1.9

1 0 . 1 Calculate the percentage of the total catch that was by-catch.

[2 marks]

Answer ☐ %

1 0 . 2 Explain why shrimp trawling produces such a high proportion of by-catch.

[1 mark]

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1 0 . 3 Explain why k-selected species such as the common skate and the Greenland shark are especially vulnerable to overfishing.

[3 marks]

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1 0 . 4 Evaluate the methods that can be used to reduce the environmental impacts of fishing.

[9 marks]

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Write an essay on one of the following topics.

11 . 1 Discuss the extent to which agriculture can be made sustainable while feeding a growing human population.

[25 marks]

OR

11 . 2 Using named examples, discuss how the ecological features of a habitat determine its importance, the threats it faces and the conservation strategies that are appropriate.

[25 marks]

Indicate which question you have answered by writing its number, **11 . 1** or **11 . 2**, at the top of your answer.

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END OF QUESTIONS